

Lab-3.1

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Lab 3: Advanced Prompt Engineering · Zero-shot, One-shot, and Few-shot Techniques

Lab Objectives:

- To explore and apply different levels of prompt examples in AI-assisted code generation.
- To understand how zero-shot, one-shot, and few-shot prompting affect AI output quality.
- To evaluate the impact of context richness and example quantity on AI performance.

Sample Data:

No.	Chatbot User Query	Intent
Q1	I forgot my password and cannot access my account.	Account Issue
Q2	Where is my order?	Order Status
Q3	Does this laptop have 16 GB RAM?	Product Inquiry
Q4	What are your customer service hours?	General Question
Q5	My account has been locked after several login attempts.	Account Issue

Zero-shot Prompting:

Prompt Used:

Zero-shot Prompt:

Classify the intent of the customer query:

"Where is my order?"

Categories: [Account Issue, Order Status, Product Inquiry, General Question]

Output:

Intent: Order Status

Observation: The model correctly classified the email without using any examples.

One-shot Prompting:

Prompt Used:

One-shot Prompt:

Classify customer query into categories.

Example:

Query: "I forgot my password and cannot access my account." -> Intent: Account Issue

Query: "Does this laptop have 16 GB RAM?" -> Intent:

Output:

Intent: Product Inquiry

Observation: Providing one labeled example helped the model understand the expected classification format and category meanings.

Few-shot Prompting:

Prompt Used:

Few-shot Prompt:

Classify customer queries:

- "Where is my order?" -> Intent: Order Status
- "What are your customer service hours?" -> Intent: General Question
- "My account has been locked after several login attempts." -> Intent: Account Issue

Query: "I need to return my package" -> Intent:

Output:

Intent: Order Status / Return Request

Observation: Few-shot prompting provided multiple examples covering different categories, making the classification more consistent and accurate.

Comparison of Zero-shot, One-shot, and Few-shot Prompting

Prompting Method	Examples Used	Output	Observation
Zero-shot	0	Order Status	Correct classification without using any examples.
One-shot	1	Feedback / Product Inquiry	One labeled example improved clarity and helped the model understand the expected classification format.
Few-shot	3-4	Others / Order Status	Multiple examples covering different categories made the classification more consistent and accurate.